

Sensorless Control of PMSM Drives Using Octant Newton–Raphson Method and Reduced-Order EKF Considering Speed Reversal and Noise

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Abstract—For back electromotive force-based sensorless control in permanent magnet synchronous motor (PMSM) drives, the conventional phase-locked loop (PLL) exhibits a $\pm\pi$ position estimation error during speed reversal. This article presents an octant Newton–Raphson method (ONRM)-based PLL, which employs a newly defined objective function and a position octant method. It not only ensures the position estimation converges correctly during speed reversal but also improves dynamic performance compared to the conventional PLL. The applicability condition of the ONRM in the presence of noise is also analyzed in detail. Subsequently, due to the noise sensitivity of the ONRM, a reduced-order extended Kalman filter is designed to filter out the nonGaussian noise in the position estimation while simultaneously estimating the speed. The effectiveness of the proposed sensorless control scheme is verified on a 1.5 kW PMSM drive platform.

Index Terms—Extended Kalman filter (EKF), Newton–Raphson method (NRM), phase-locked loop (PLL), sensorless control, speed reversal.

I. INTRODUCTION

SINCE the high efficiency and high-power density, permanent-magnet synchronous motors (PMSMs) are extensively used in electric vehicles and industrial fields [1], [2], [3]. For a high-performance PMSM control system, precise rotor position and speed information are essential. However, mechanical sensors increase system costs, volume, complexity, and unreliability. Consequently, sensorless control techniques have been favored and widely studied by researchers.

Sensorless control refers to deriving the rotor speed and position information solely through the detection of electrical signals in the motor windings without mechanical position sensors. The sensorless control techniques based on the fundamental mathematical model depend on the rotational position-related quantities to estimate rotor position and speed. Currently,

commonly used algorithms include sliding mode observer algorithm [4], model reference adaptive control algorithm [5], extended Kalman filter (EKF) algorithm [6], etc. For techniques based on back electromotive force (EMF) [7], [8], a position extraction algorithm is required after estimating the back EMF. The arctangent method is the most straightforward, but it is susceptible to noise and harmonics [9].

To extract accurate rotor position and speed from the estimated back EMF, the phase-locked loop (PLL) is intensively studied for sensorless control techniques [9], [10], [11], [12], [13]. It has a simple structure and is easy to apply. However, it also has several limitations, including unsatisfactory dynamic and steady-state performance, laborious tuning, and difficulty in adapting to changing operating conditions [11]. In addition, in most literature, the PLL is simplified to a linear system under the assumption of a small tracking error [9], [10], [11], [12], ignoring the nonlinearity in the dynamic process. In fact, due to the nonlinear characteristics of the PLL, cycle slipping may occur under certain conditions, significantly increasing the convergence time of the estimation [13], [14]. Recently, a novel category of PLL methods has been proposed, which is based on an objective function and iterative search [15], [16], [17], [18]. They do not need dynamic processes, convergence time, or parameter tuning, and can avoid performance degradation caused by nonlinear characteristics, such as cycle slipping. Therefore, they naturally hold advantages over conventional PLL, especially in scenarios emphasizing rapid position tracking. The finite position set-PLL is first introduced in [15] by leveraging the concept of finite-control-set model predictive control. It discretizes the position into several finite states and minimizes the objective function across all possible positions. Nevertheless, the traversing method results in a large number of iterations and a high computational load. A substantial improvement in accuracy is achieved with just three iterations by employing the Newton–Raphson method (NRM) [18], which satisfies the accuracy needed for most operations. A position quartering method is also presented to set initial values, securing the convergence of NRM. The computational load is further reduced in [19] and [20].

However, for sensorless control techniques based on back EMF, speed reversal and noise are critical issues to be considered for better estimation performance. Since the sign of the back EMF expression is related to the direction of rotation, the estimated position will differ from the actual value by π

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during speed reversal. In [18] and [21], the sign of the speed estimation is directly utilized to distinguish forward rotation from reverse rotation, followed by corresponding compensation. But this method is hard to realize online and difficult to prove its effectiveness theoretically. Therefore, several methods specifically addressing the speed reversal issue have been proposed, the core principle of which is to eliminate the influence of the speed sign on the estimation error. In [22], the position error equation is constructed based on the tangent function and later extended to a more concise form in [23] and a more general form in [24]. However, the analysis in both studies is approximated as a linear system and does not fully account for the dynamics introduced by the tangent function. In [25], the phase detector in the PLL is designed using the double-angle formula, making the algorithm more intuitive. A dual-PLL approach is proposed in [9] by increasing the order of the PLL, but the number of parameters that need to be tuned also increases to four. Furthermore, all these methods lack a rigorous analysis of their applicability conditions. In fact, their effectiveness may not be guaranteed in the presence of cycle slipping.

The calculation of back EMF using the discrete voltage equations of the PMSM retains the noise in the estimations [15], [16], [17], [18], thereby greatly reducing the performance of sensorless control. The estimated speed can be filtered using low pass filtering [15], [16] or mean filtering [17], [18], but directly filtering the estimated position results in a large phase delay, which is unacceptable. A third-order extended state observer is proposed in [11], which enhances noise suppression capability and maintains the observation bandwidth, but its parameter-tuning process is relatively complex. In [12], an adaptive vectorial filter is applied to filter the estimated back EMF, indirectly reducing noise in the estimated position. However, the effectiveness of this filter is highly dependent on the accuracy of the speed estimation. It causes a coupling between position estimation error and speed estimation error, and increases the complexity of system analysis. A simplified EKF is introduced in [26], which simplifies the computation of the Kalman gain but at the cost of reduced filtering performance to some extent. It is worth noting that none of the above methods take advantage of the noise distribution information, and they are mainly effective for high-frequency noise.

The main contribution of this article is to resolve the issues of speed reversal and noise sensitivity in sensorless control techniques based on back EMF. First, an octant NRM (ONRM) for position estimation is introduced to avoid errors during speed reversal by integrating the proposed position octant method with the NRM. Then, a well-defined applicability condition of the ONRM in the presence of noise is analyzed and simplified into a constraint on the speed range, making it easier to apply in practice. Finally, due to Gaussian measurement noise causing nonGaussian noise in the position estimation, a reduced-order extended Kalman filter (REKF) is implemented for filtering and speed estimation. The noise is converted to Gaussian by a special design of the measurement equation. It ensures noise resistance and reasonable computational complexity.

The rest of this article is organized as follows. Section II presents the basic principles of sensorless control and the conventional PLL. Section III details the proposed ONRM and

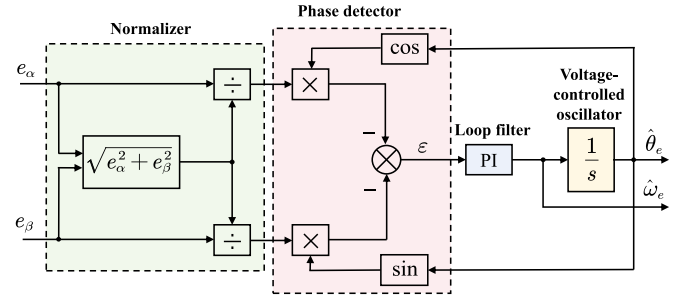


Fig. 1. Block diagram of the conventional PLL.

REKF, respectively. The experimental results are discussed in Section IV. Finally, Section V concludes this article.

II. MODEL AND PROBLEM FORMULATION

A. Mathematical Model of SPMSM

The mathematical model of a three-phase surface-mounted PMSM (SPMSM) system in the stationary $\alpha\beta$ reference frame can be expressed as follows:

$$\begin{cases} u_\alpha = R_s i_\alpha + L_s \frac{di_\alpha}{dt} + e_\alpha \\ u_\beta = R_s i_\beta + L_s \frac{di_\beta}{dt} + e_\beta \end{cases} \quad (1)$$

where $i_\alpha, i_\beta, u_\alpha, u_\beta, e_\alpha$, and e_β are the $\alpha\beta$ -axes stator currents, stator voltages, flux linkage, and back EMF, respectively. R_s is the stator resistance, and L_s is the stator inductance. The expression of back EMF can be described as follows:

$$\begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} = \omega_e \varphi_f \begin{bmatrix} -\sin \theta_e \\ \cos \theta_e \end{bmatrix} \quad (2)$$

where φ_f is the permanent-magnet flux linkage; $\theta_e = n_p \theta_r$ and $\omega_e = n_p \omega_r$ are the electrical rotor position and the electrical rotor speed, respectively; θ_r and ω_r are the mechanical rotor position and the mechanical rotor speed, respectively; and n_p is the number of pole pairs. It can be noted that the back EMF in (2) contains the position θ_e . Therefore, different observers can be constructed to estimate the back EMF [1].

For convenience, the backward Euler method [12], [27] is used to discretize (1) and directly calculate the back EMF as follows:

$$\begin{cases} \hat{e}_\alpha(k) = u_\alpha(k) - R_s i_\alpha(k) - L_s \frac{i_\alpha(k) - i_\alpha(k-1)}{T_s} \\ \hat{e}_\beta(k) = u_\beta(k) - R_s i_\beta(k) - L_s \frac{i_\beta(k) - i_\beta(k-1)}{T_s} \end{cases} \quad (3)$$

where T_s is the sampling period. This calculation does not require speed information, resulting in the decoupling of position estimation from speed estimation and simplifying system structure.

B. Conventional PLL

After obtaining the back EMF in the $\alpha\beta$ axis, the PLL is generally adopted to estimate the speed and position [4]. The conventional PLL includes four modules: amplitude normalizer, phase detector, loop filter, and voltage-controlled oscillator, as shown in Fig. 1. The speed estimation is achieved through the PI-type loop filter, followed by an integration stage to estimate

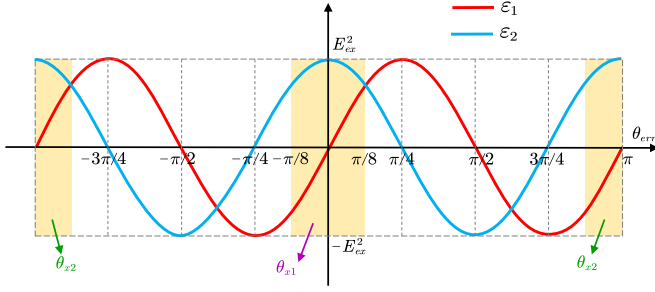


Fig. 6. Relationship between ε_1 , ε_2 , and θ_{err} .

the nonGaussian noise in the position estimation and estimate the speed simultaneously. In addition, a detailed analysis of the applicability condition is provided.

A. Position Octant Method and NRM

Assuming the motor is operating in forward rotation, the equivalent position error indexes ε_1 , ε_2 , and the cost function J are defined as follows:

$$\begin{aligned}\varepsilon_1 &= -2e_\alpha e_\beta \cos 2\hat{\theta}_e + (e_\alpha^2 - e_\beta^2) \sin 2\hat{\theta}_e \\ &= 2E_{ex}^2 \sin \theta_e \cos \theta_e \cos 2\hat{\theta}_e \\ &\quad + E_{ex}^2 (\sin^2 \theta_e - \cos^2 \theta_e) \sin 2\hat{\theta}_e \\ &= E_{ex}^2 \sin (2\theta_{\text{err}})\end{aligned}\quad (8)$$

$$\begin{aligned}\varepsilon_2 &= (e_\beta^2 - e_\alpha^2) \cos 2\hat{\theta}_e - 2e_\alpha e_\beta \sin 2\hat{\theta}_e \\ &= E_{ex}^2 (\cos^2 \theta_e - \sin^2 \theta_e) \cos 2\hat{\theta}_e \\ &\quad + 2E_{ex}^2 \sin \theta_e \cos \theta_e \sin 2\hat{\theta}_e \\ &= E_{ex}^2 \cos (2\theta_{\text{err}})\end{aligned}\quad (9)$$

$$J = |\varepsilon_1| \quad (10)$$

where $E_{ex} = \sqrt{e_\alpha^2 \cdot e_\beta^2} = |\omega_e \varphi_f|$ is the amplitude of the back EMF. Equations (8) and (9) remain the same during both forward and reverse operations, thus the problem of speed reversal is avoided. Fig. 6 illustrates the relationship between ε_1 , ε_2 , and θ_{err} . The following objective is to find the $\hat{\theta}_e$ that minimizes the cost function J . To select an appropriate initial value for the iteration, this article proposes the position octant method.

Initially, eight evenly distributed prospective initial values are generated based on $\hat{\theta}_e(k-1)$, which is the position from the previous moment. The range of the angle values is $(-\pi, \pi]$.

$$\begin{cases} \theta_{-3\pi/4} = \hat{\theta}_e(k-1) - 3\pi/4 \\ \theta_{-2\pi/4} = \hat{\theta}_e(k-1) - 2\pi/4 \\ \theta_{-\pi/4} = \hat{\theta}_e(k-1) - \pi/4 \\ \theta_0 = \hat{\theta}_e(k-1) + 0 \\ \theta_{\pi/4} = \hat{\theta}_e(k-1) + \pi/4 \\ \theta_{2\pi/4} = \hat{\theta}_e(k-1) + 2\pi/4 \\ \theta_{3\pi/4} = \hat{\theta}_e(k-1) + 3\pi/4 \\ \theta_\pi = \hat{\theta}_e(k-1) + \pi \end{cases} \quad (11)$$

Subsequently, based on the conditions $\varepsilon_2 > 0$ and $\min J$, two initial values can be selected, tentatively called θ_{x1} and θ_{x2} . $|\theta_{x1} - \theta_{x2}| = \pi$, and their corresponding θ_{err} must lie within the colored regions of Fig. 6. Assume that θ_{x1} corresponds to the interval $(-\pi/8, +\pi/8)$, and θ_{x2} corresponds to the interval $(-\pi, -7\pi/8) \cup (7\pi/8, \pi]$. If the initial value is θ_{x1} , θ_{err} will converge to $2n\pi$. If the initial value is θ_{x2} , θ_{err} will converge to $(2n+1)\pi$. Due to the difficulty in distinguishing between θ_{x1} and θ_{x2} , the following applicability condition is presented:

$$|\theta_e(k) - \hat{\theta}_e(k-1)| < 3\pi/8. \quad (12)$$

This implies that the estimated position at time $k-1$ should not differ too much from the actual position at time k . In this scenario, θ_{x1} necessarily takes one of $\theta_{-\pi/4}$, θ_0 , or $\theta_{\pi/4}$, and θ_{x2} takes one of $\theta_{3\pi/4}$, θ_π , or $\theta_{-3\pi/4}$. Thus, in practical computation, it suffices to calculate and verify the values of $\theta_{-\pi/4}$, θ_0 , and $\theta_{\pi/4}$ to determine θ_{x1} . A detailed analysis and a practical approach for judging (12) are provided in the next section. In addition, the approximate range of the actual position at $k=0$ is required to be known.

After obtaining the initial value θ_{x1} , the NRM can be used to solve the following equation to obtain $\hat{\theta}_e$:

$$\varepsilon_1(\hat{\theta}_e) = E_{ex}^2 \sin [2(\theta_e - \hat{\theta}_e)] = 0. \quad (13)$$

Through derivation, the iterative formula can be obtained as follows:

$$\hat{\theta}_{e(j)} = g(\hat{\theta}_{e(j-1)}) = \hat{\theta}_{e(j-1)} + \frac{\varepsilon_1(\hat{\theta}_{e(j-1)})}{2\varepsilon_2(\hat{\theta}_{e(j-1)})} \quad (14)$$

where j is the number of iterations. The sufficient conditions for the astringency of NRM are as follows: 1) The derivative of ε_1 with respect to $\hat{\theta}_e$ is not equal to zero; 2) The second derivative of ε_1 with respect to $\hat{\theta}_e$ exists; 3) The derivative of $g(\hat{\theta}_e)$ with respect to $\hat{\theta}_e$ is less than 1. It can be calculated as follows:

$$\begin{aligned}\frac{d\varepsilon_1}{d\hat{\theta}_e} &= -2E_{ex}^2 \cos [2(\theta_e - \hat{\theta}_e)] \\ &= -2E_{ex}^2 \cos (2\theta_{\text{err}})\end{aligned}\quad (15)$$

$$\begin{aligned}\frac{d^2\varepsilon_1}{d\hat{\theta}_e^2} &= 2E_{ex}^2 \sin [2(\theta_e - \hat{\theta}_e)] \\ &= 2E_{ex}^2 \sin (2\theta_{\text{err}})\end{aligned}\quad (16)$$

$$\begin{aligned}\left| \frac{dg(\hat{\theta}_e)}{d\hat{\theta}_e} \right| &= \left| 1 - \frac{\cos^2(2\theta_{\text{err}}) - \sin^2(2\theta_{\text{err}})}{\cos^2(2\theta_{\text{err}})} \right| \\ &= \frac{\sin^2(2\theta_{\text{err}})}{\cos^2(2\theta_{\text{err}})} \\ &= \tan^2(2\theta_{\text{err}}).\end{aligned}\quad (17)$$

Due to the application of the position octant method, the range of θ_{err} has been reduced to the interval $(-\pi/8, \pi/8)$. Therefore, it is evident that these three conditions can be satisfied within this interval.

TABLE I
 THEORETICAL ESTIMATION ERROR OF THE ONRM

$\theta_{err(0)}$ (rad)	$\theta_{err(2)}$ (rad)	$\theta_{err(3)}$ (rad)	$\theta_{err(4)}$ (rad)
$\pi/8$	1.68e-3	-6.30e-9	3.34e-25
$\pi/10$	1.59e-4	-5.31e-12	2.00e-34
$\pi/12$	2.59e-5	-2.32e-14	1.67e-41
0	0	0	0
$-\pi/13$	-1.19e-5	2.27e-15	-1.55e-44
$-\pi/11$	-6.09e-5	3.02e-13	-3.67e-38
$-\pi/9$	-4.71e-4	1.39e-10	-3.58e-30

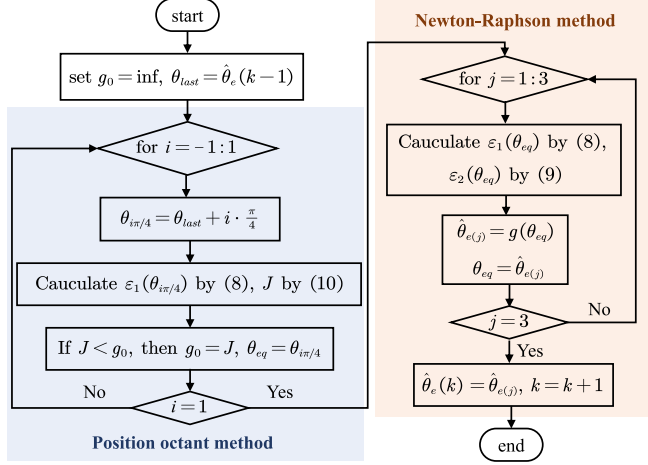


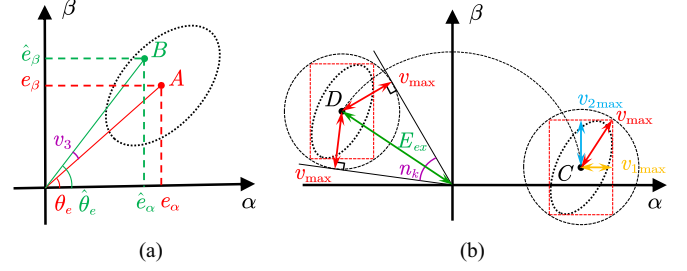
Fig. 7. Flowchart of the proposed ONRM.

Furthermore, to reasonably determine the iteration number, the expression for computation accuracy can be derived as follows:

$$\begin{aligned}
 \theta_{err(j)} &= \theta_e - \hat{\theta}_{e(j)} \\
 &= \theta_e - \left[\hat{\theta}_{e(j-1)} + \frac{E_{ex}^2 \sin(2\theta_{err(j-1)})}{2E_{ex}^2 \cos(2\theta_{err(j-1)})} \right] \\
 &= \theta_e - \left[(\theta_e - \theta_{err(j-1)}) + \frac{\sin(2\theta_{err(j-1)})}{2 \cos(2\theta_{err(j-1)})} \right] \\
 &= \theta_{err(j-1)} - \frac{\tan(2\theta_{err(j-1)})}{2}
 \end{aligned} \quad (18)$$

where $\theta_{err(j)}$ denotes the estimated error after j iterations and $\theta_{err(0)}$ is the iterative initial value. The estimated position errors after 2, 3, and 4 iterations are calculated using different initial iteration values, as shown in Table I. It can be observed that the larger the initial value, the larger the error for the same number of iterations. After three iterations, the maximum error is 6.3e-9, which can meet the precision requirements of most cases. Therefore, to balance computational load and accuracy, performing three iterations is a reasonable choice.

The flowchart of the proposed ONRM is represented in Fig. 7. It consists of two loops with indices i and j , which represent the position octant method and NRM, respectively. i ranges from -1 to 1 , selecting the correct initial value θ_{x1} from $\theta_{-\pi/4}$, θ_0 , and $\theta_{\pi/4}$. Then, j ranges from 1 to 3 , performing three iterations of


 Fig. 8. Schematic illustration in the $\alpha\beta$ -axis. (a) Distribution range of back EMF. (b) Replace the ellipse with the circumcircle of its circumscribed rectangle.

calculations. After obtaining $\hat{\theta}_e$, $\hat{\omega}_e$ can be derived by applying differentiation followed by low-pass filtering [15].

B. Applicability Condition of the ONRM Under Noise

The $\hat{e}_\alpha$ and $\hat{e}_\beta$ calculated from (3) inevitably contain significant noises, which are ultimately reflected in $\hat{\theta}_e$. The noises not only increase the position error but also complicate the assessment of the applicability condition of the ONRM, as expressed in (12). This can be expressed mathematically as follows:

$$\begin{bmatrix} \hat{e}_\alpha \\ \hat{e}_\beta \end{bmatrix} = \begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\text{sgn}(\omega_e) \hat{E}_{ex} \sin \hat{\theta}_e \\ \text{sgn}(\omega_e) \hat{E}_{ex} \cos \hat{\theta}_e \end{bmatrix} \quad (19)$$

where $\hat{\theta}_e = \theta_e + v_3$, $\hat{E}_{ex} = E_{ex} + v_4$, and $\hat{E}_{ex} = \sqrt{\hat{e}_\alpha^2 + \hat{e}_\beta^2}$. v_1, v_2, v_3 , and v_4 are the noises contained in $\hat{e}_\alpha, \hat{e}_\beta, \hat{\theta}_e$, and $\hat{E}_{ex}$, respectively. The measurement noise of current $i_{\alpha\beta}$ and voltage $u_{\alpha\beta}$ are denoted as v_{i1}, v_{i2}, v_{u1} , and v_{u2} , respectively.

Assumption 1: v_{i1}, v_{i2}, v_{u1} , and v_{u2} follow a Gaussian distribution with a mean of zero and are bounded. Furthermore, v_{i1} and v_{i2} are correlated, as are v_{u1} and v_{u2} .

According to (3), v_1 and v_2 can be expressed as follows:

$$\begin{aligned}
 v_1(k) &= v_{u1}(k) - R_s v_{i1}(k) - \frac{L_s}{T_s} [v_{i1}(k) - v_{i1}(k-1)] \\
 v_2(k) &= v_{u2}(k) - R_s v_{i2}(k) - \frac{L_s}{T_s} [v_{i2}(k) - v_{i2}(k-1)].
 \end{aligned} \quad (20)$$

Considering the closure property of the Gaussian distribution under linear transformations, it can be concluded that v_1 and v_2 follow a bivariate Gaussian distribution with a mean of zero and a correlation coefficient ρ . v_1 and v_2 are also bounded, and the maximum values of them can be denoted as $v_{1\max}$ and $v_{2\max}$, respectively. Assuming $\rho > 0$, Fig. 8(a) illustrates this distribution in the $\alpha\beta$ -coordinate system, where point A represents the actual back EMF, and point B represents the estimated back EMF affected by noise. The dashed line, centered at point A, represents the range of noise distribution. Due to the presence of ρ , the shape of the dashed line is an ellipse.

Based on the above analysis, the condition (12) can be rewritten as follows:

$$|\theta_e(k) - \theta_e(k-1) - v_3(k-1)| < 3\pi/8. \quad (21)$$

According to the triangle inequality, it becomes

$$|v_3(k-1)| < 3\pi/8 - |\omega_e T_s|. \quad (22)$$

It is complex to calculate the angle n_k between the two tangents drawn from the origin to the ellipse. In addition, n_k varies with changes in θ_e when E_{ex} remains constant. To simplify derivation and computation, the ellipse is replaced by the circumcircle of its circumscribed rectangle. As shown in Fig. 8(b), the radius of the circumcircle is $v_{\max} = \sqrt{v_{1\max}^2 + v_{2\max}^2}$. By simple geometric calculations, (22) becomes

$$n_k = 2 \arcsin(v_{\max}/E_{ex}) < 3\pi/4 - 2|\omega_e T_s|. \quad (23)$$

Although the actual speed is difficult to obtain, a conservative estimate can be made by using the rated speed or maximum speed, i.e.,

$$v_{\max} < E_{ex} \sin(3\pi/8 - |\omega_{\max} T_s|). \quad (24)$$

Substituting $E_{ex} = |\omega_e| \varphi_f$, it can be obtained that

$$|\omega_e| > \frac{v_{\max}}{\varphi_f \sin(3\pi/8 - |\omega_{\max} T_s|)}. \quad (25)$$

For example, by taking $\omega_{\max} = 600$ rad/s and $T_s = 1\text{e-}4\text{s}$, (24) can be calculated as follows:

$$v_{\max} < 0.899 E_{ex}. \quad (26)$$

It implies that the condition (12) is satisfied when the maximum of the noise is less than 0.899 times the amplitude of the back EMF. If the flux linkage φ_f and $v_{\max}$ are known, the applicable speed range for ONRM can be calculated by (25) as follows:

$$|\omega_e| > v_{\max} / (0.899 \varphi_f) \quad (27)$$

which is meaningful in practical applications.

C. REKF for Filtering and Speed Estimation

According to (19), it can be seen that v_1 and v_2 represent the noise in the Cartesian coordinate system, while v_3 and v_4 represent the noise in the polar coordinate system, i.e.,

$$v_3 = \arctan\left(\frac{v_2}{v_1}\right), v_4 = \sqrt{v_1^2 + v_2^2}. \quad (28)$$

It should be noted that $\hat{E}_{ex}$ can always be eliminated through the normalization process, we should focus more on v_3 in $\hat{\theta}_e$. Since the ONRM lacks noise suppression capability, it is necessary to filter $\hat{\theta}_e$ to reduce the influence of v_3 . However, v_3 is nonGaussian, but follows a Beckmann distribution [28]. The probability density function of v_3 is given in [29]. Fig. 9 illustrates the distribution of v_3 . It can be observed that when $\rho \neq 0$, the distribution is not symmetric and varies for different θ_e values.

To avoid directly filtering nonGaussian noise and estimate the speed simultaneously, a REKF is designed based on the following system:

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{w} \\ \mathbf{y} = \mathbf{h}(\mathbf{x}) + \mathbf{v}. \end{cases} \quad (29)$$

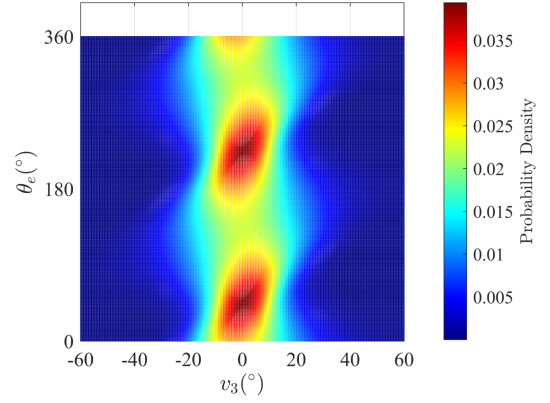


Fig. 9. Probability density curves of v_3 for different θ_e values with $\rho = 0.5$.

where

$$\begin{aligned} \mathbf{x} &= \begin{bmatrix} \theta_e \\ \omega_e \end{bmatrix}, \mathbf{f}(\mathbf{x}) = \begin{bmatrix} \omega_e \\ 0 \end{bmatrix}, \mathbf{h}(\mathbf{x}) = |\omega_e| \varphi_f \begin{bmatrix} \sin \theta_e \\ \cos \theta_e \end{bmatrix} \\ \mathbf{w} &= \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -\text{sgn}(\omega_e) v_1 \\ \text{sgn}(\omega_e) v_2 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} \hat{E}_{ex} \sin \hat{\theta}_e \\ \hat{E}_{ex} \cos \hat{\theta}_e \end{bmatrix}. \end{aligned} \quad (30)$$

The measurement part uses the form of back EMF without the sign. This ensures the measurement noise $\mathbf{v}$ is Gaussian, and preserves the speed reversal characteristics in the ONRM. Reduced order refers to the fact that compared to the fourth-order or third-order EKF in [6] and [30], the REKF designed by (29) is only second-order. This requires fewer computations and has fewer noise covariance matrix elements facilitating parameter tuning. The Jacobian matrices are defined as follows:

$$\begin{aligned} \mathbf{F} &= \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ \mathbf{H} &= \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \\ &= \begin{bmatrix} |\omega_e| \varphi_f \cos \theta_e & \text{sgn}(\omega_e) \varphi_f \sin \theta_e \\ -|\omega_e| \varphi_f \sin \theta_e & \text{sgn}(\omega_e) \varphi_f \cos \theta_e \end{bmatrix}. \end{aligned} \quad (31)$$

It can be observed that $\mathbf{F}$ is a constant matrix. The discrete form of (29) can be written as follows:

$$\begin{cases} \mathbf{x}_k = (\mathbf{I} + \mathbf{F}T_s) \mathbf{x}_{k-1} + \mathbf{p}_{k-1} \\ \mathbf{y}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{q}_k \end{cases} \quad (32)$$

where $\mathbf{p}_k$ and $\mathbf{q}_k$ are the discrete forms of $\mathbf{w}$ and $\mathbf{v}$. The steps of the REKF can be described by the following:

1) Prediction:

$$\hat{\mathbf{x}}_{k|k-1} = (\mathbf{I} + \mathbf{F}T_s) \hat{\mathbf{x}}_{k-1} \quad (33)$$

$$\mathbf{P}_{k|k-1} = (\mathbf{I} + \mathbf{F}T_s) \mathbf{P}_{k-1} (\mathbf{I} + \mathbf{F}T_s)^T + \mathbf{Q} \quad (34)$$

2) Updating:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T \left(\mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T + \mathbf{R} \right)^{-1} \quad (35)$$

TABLE II
 MAIN PARAMETERS OF SPMSM

Parameter	Symbol	Value
Pole pairs	n_p	4
Stator resistance	R_s	2.16 Ω
Stator inductance	L_s	18.7 mH
Permanent-magnet flux linkage	φ_f	0.32 Wb
Inertia	J	0.0019 kg · m ²
Rated speed	N_r	1500 rpm
Rated power	P_r	1.5 kW
Rated torque	T_r	10 N·m
Rated voltage	V_r	380 V
Rated current	I_r	4 A

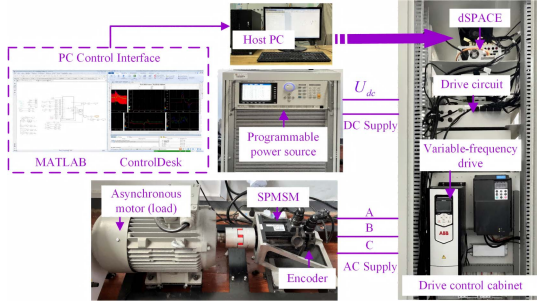


Fig. 10. Experimental setup.

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{y}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1})) \quad (36)$$

$$\mathbf{P}_k = \mathbf{P}_{k|k-1} - \mathbf{K}_k \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \quad (37)$$

where $\mathbf{P}$ is the error covariance matrix; $\mathbf{K}$ is the Kalman gain; $\mathbf{Q}$ and $\mathbf{R}$ are the noise covariance matrices. In addition, the initial value $\hat{\theta}_{e0}$ from the ONRM at the starting moment can be used as the initial state θ_{e0} in the REKF, thereby reducing the convergence time.

IV. EXPERIMENTAL RESULTS

To verify the effectiveness of the proposed sensorless control schemes, experimental tests are conducted on a 1.5 kW SPMSM system. The parameters of the SPMSM are shown in Table II. The experimental setup is shown in Fig. 10. A dSPACE DS1202 system is used to carry out the real-time algorithm. An asynchronous motor driven by ABB's variable-frequency drive provides the load torque for the SPMSM. In the experiments, both the sampling and switching frequency of the drive are set to 10 kHz. According to the pole placement method [13], parameters of the PI controller in conventional PLL are set to $k_p = 200$ and $k_i = 10000$. The parameters in the REKF were tuned and set to $\mathbf{Q} = [1 \ 0; 0 \ 5000]$ and $\mathbf{R} = [2000 \ 0; 0 \ 2000]$ to achieve better performance. Except for Fig. 14, all experiments are conducted under closed-loop conditions, where the estimated position and speed are fed back to the system, i.e., the motor operates in sensorless mode.

A. Comparison Under Steady-State and Noisy Conditions

In this section, the motor is operated at 1500 rpm separately using three methods: conventional PLL, ONRM, and ONRM-REKF, to evaluate their position and speed estimation

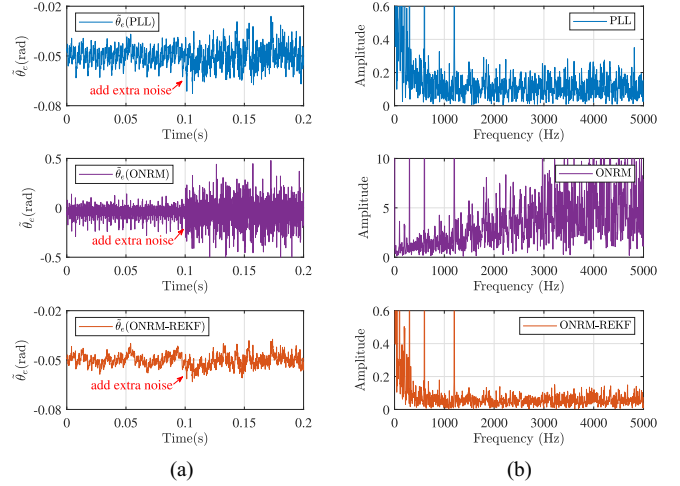


Fig. 11. Experimental results of position estimation error under steady-state and noisy conditions. (a) Position estimation error over time. (b) Spectral analysis.

accuracy under steady-state conditions. Subsequently, noise with a correlation coefficient of 0.5 and a variance of 0.08 is added to the current measurements i_α and i_β at $t = 0.1$ s to validate the performance under noisy conditions. The error between the actual and estimated electrical position is represented as $\tilde{\theta}_e = \theta_e - \hat{\theta}_e$, and the error between the actual and estimated rotor speeds is represented as $\tilde{\omega}_r = \omega_r - \hat{\omega}_r$. Fig. 11(a) presents the position estimation error throughout the entire process for the three methods. It can be observed that the estimation accuracy of all three methods decreases after the introduction of additional noise. Among them, the ONRM is the most affected by noise, exhibiting the largest fluctuation range in position estimation error. The fluctuation range of the position estimation error for the ONRM-REKF is slightly smaller than that of the conventional PLL under both steady-state and noisy conditions. And their average position estimation errors are almost identical, both approximately 0.05 rad. Fig. 11(b) shows the spectral analysis results for the data in the time interval 0.1 s–0.2 s from Fig. 11(a). It represents the magnitude intensity at different frequencies. From the frequency domain perspective, it is evident that the proposed ONRM-REKF demonstrates good noise reduction capability. Fig. 12 shows the corresponding speed estimation error throughout the same process. It can be seen that the fluctuation of the speed estimation error for the ONRM-REKF is significantly smaller than those of the conventional PLL and the ONRM. The results of the frequency domain analysis further confirm this conclusion.

A quantitative analysis of this experiment is provided in Table III. The mean absolute error (MAE) and the standard deviation of the error (STDE) are defined as follows:

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N |e(k)|, \text{STDE} = \sqrt{\frac{1}{N} \sum_{k=1}^N (e(k) - \text{MAE})^2} \quad (38)$$

where e is the estimation error $\tilde{\theta}_e$ or $\tilde{\omega}_r$, k is the sampling instant, and N is the number of sampling points. These metrics

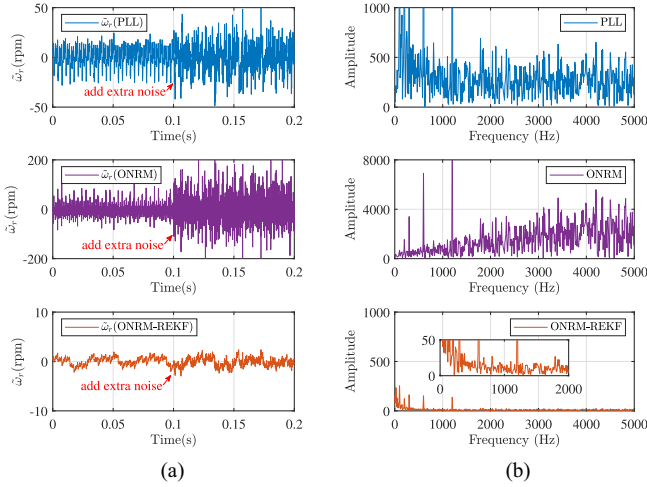


Fig. 12. Experimental results of speed estimation error under steady-state and noisy conditions. (a) Speed estimation error over time. (b) Spectral analysis.

TABLE III
PERFORMANCE COMPARISON BETWEEN THREE SENSORLESS
CONTROL METHODS

Performance Indexes	PLL	ONRM	ONRM -REKF	Improvement over PLL (%)
SS: MAE of $\hat{\theta}_e$	0.050	0.0599	0.0499	0.2
SS: MAE of $\hat{\omega}_r$	7.964	19.97	0.6494	91.6
SS: STDE of $\hat{\theta}_e$	0.0047	0.0586	0.0030	36.2
SS: STDE of $\hat{\omega}_r$	9.980	26.12	0.7717	92.3
SSN: MAE of $\hat{\theta}_e$	0.0505	0.1132	0.0503	0.4
SSN: MAE of $\hat{\omega}_r$	10.93	47.33	0.7244	93.4
SSN: STDE of $\hat{\theta}_e$	0.0068	0.1394	0.0040	41.2
SSN: STDE of $\hat{\omega}_r$	14.16	62.11	0.8736	93.8
Execution time(μ s)	0.62	1.16	2.86	-361.3

Notes: SS: Steady-state (at 0s-0.1s in Fig. 11(a) and Fig. 12(a)); SSN: Steady-state under noise (at 0.1s-0.2s in Fig. 11(a) and Fig. 12(a)).

indicate that the proposed ONRM-REKF has better steady-state performance and noise resistance in both position and speed estimation. It can be further observed that the position estimation error is larger than the error analyzed in Section III, due to the nonmodel dynamics of the machine, inverter nonlinearities, and harmonics in the back EMF. Nevertheless, the accuracy of the proposed method is still good and acceptable.

To compare the computational burden of the three methods, their execution times are measured on this platform. The execution times of conventional PLL, ONRM, and ONRM-REKF were 0.62 μ s, 1.16 μ s, and 2.86 μ s, respectively. It can be observed that the execution time of the proposed ONRM-REKF is more than four times that of the PLL. Considering the performance improvement, this level of computational burden increase is acceptable.

B. Comparison Under Speed Reversal Condition

Two experiments are designed in this section to provide a detailed comparison of the performance of the conventional PLL and the ONRM-REKF during the speed reversal process.

1) The motor is stabilized at 500 rpm and the sensorless algorithm is operated. The reference speed decreases uniformly

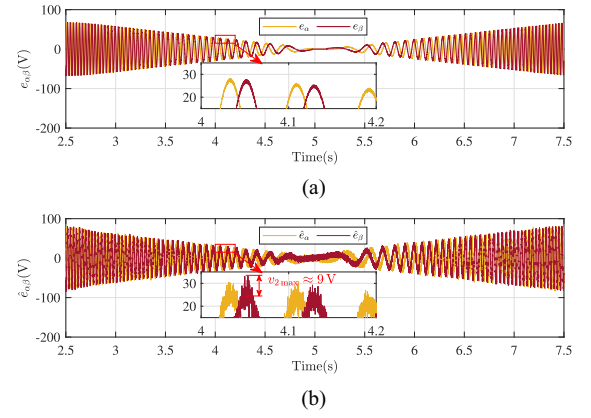


Fig. 13. Experimental results of the back EMF during the speed reversal (from +500 rpm to -500 rpm). (a) Actual back EMF. (b) Estimated back EMF.

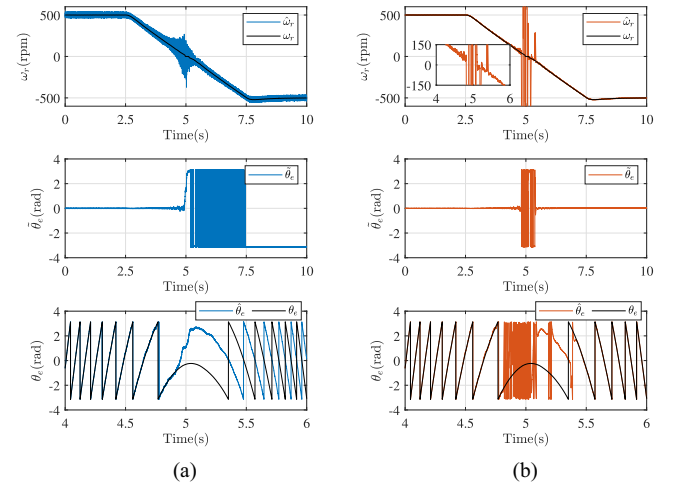


Fig. 14. Experimental results under speed reversal conditions (from +500 rpm to -500 rpm). (a) Conventional PLL. (b) ONRM-REKF.

from 500 rpm to -500 rpm between $t = 2.5$ s and $t = 7.5$ s. Fig. 13 shows the actual and estimated back EMF values e_{α} and e_{β} when the speed crosses zero. As the speed decreases, the back EMF is gradually drowned out by noise. From Fig. 13(b), it can be observed that $v_{1\max}$ and $v_{2\max}$ are approximately 9 V. Thus, $v_{\max}$ is 13 V. According to (23), taking the rated speed as the maximum speed, the corresponding applicable speed range can be conservatively estimated as $|\omega_e| > 45.3$ rad/s, i.e., $|\omega_r| > 108$ rpm. To compare the estimation errors during motor reversal, the estimated position and speed are not fed back to the system. Fig. 14 illustrates the estimation errors of speed and position for the conventional PLL and the ONRM-REKF in this case. It can be observed that the PLL becomes unstable at 42 rpm, and after the motor reverses, it converges to $-\pi$ (as analyzed in Section II). By contrast, the ONRM-REKF still converges to 0, demonstrating its robustness to speed reversal. Furthermore, the unstable speed range for the ONRM-REKF is -77 to 70 rpm, which aligns with the estimated results of (25).

2) To compare the performance at different speeds, the motor is gradually operated from 1500 rpm to -1500 rpm under no-load conditions. The trend of the speed reference is

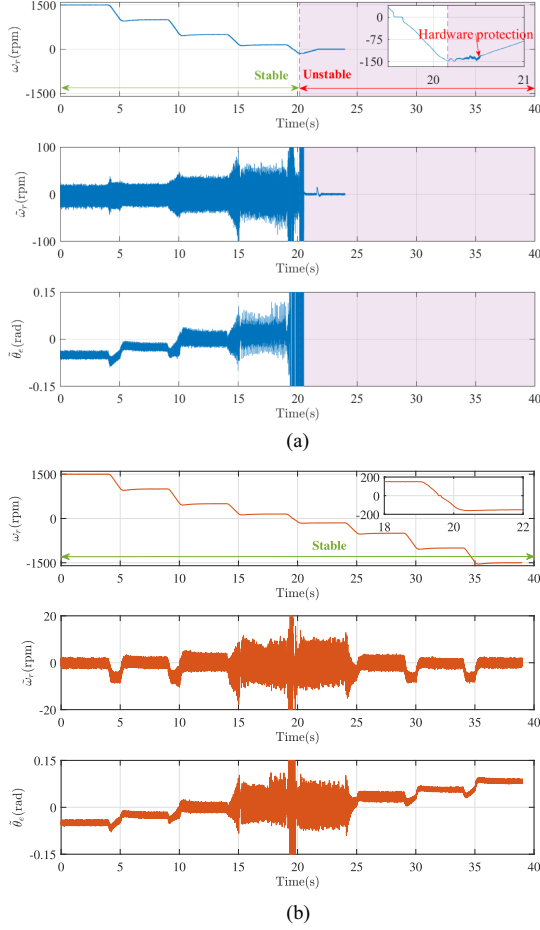


Fig. 15. Experimental results under sequentially changing speed reference conditions (from +1500 rpm to −1500 rpm). (a) Conventional PLL. (b) ONRM-REKF.

sequentially set to 1500, 1000, 500, 150, −150, −500, −1000, and −1500 rpm. The low-speed transition process from 150 rpm to −150 rpm adopts the transition strategy proposed in [31]. The actual rotor speed, speed estimation error, and position estimation error with PLL and ONRM-REKF are presented in Fig. 15. It can be observed that for the PLL, incorrect position estimation causes the whole control system to become unstable at 20.2 s, almost immediately after the transition process ends. Subsequently, at 20.5 s, the erroneous control leads to an excessive current, which triggers the hardware protection mechanism of the system. As a result, the motor stops operating, and the speed gradually decreases to zero. In contrast, the ONRM-REKF remains stable throughout the entire process. Although its position estimation error varies under different speeds, it stays within a relatively small range and does not significantly affect the performance of the speed control.

C. Comparison Under Other Operation

The actual rotor speed, electromagnetic torque T_e , speed estimation error, and position estimation error of the ONRM-REKF are presented in Fig. 16, where the load experiences several step changes and the reference speed is −500 rpm. T_e is calculated by $T_e = 1.5n_p\varphi_f i_q$. The load is sequentially set to −7, −4, −1, and

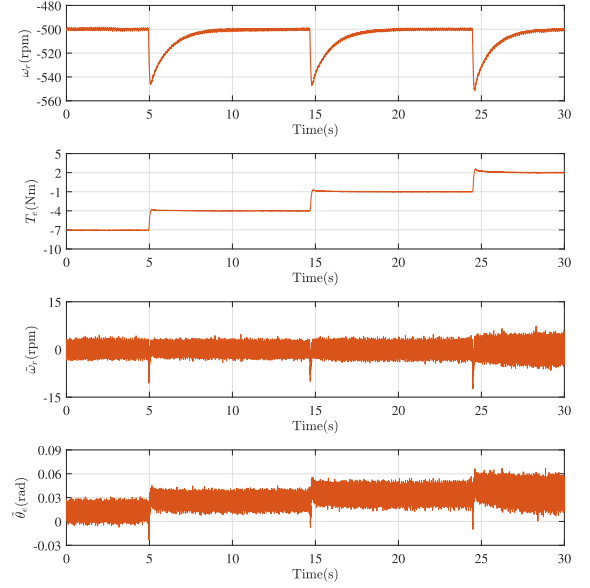


Fig. 16. Experimental results of the ONRM-REKF under step load change condition at −500 rpm.

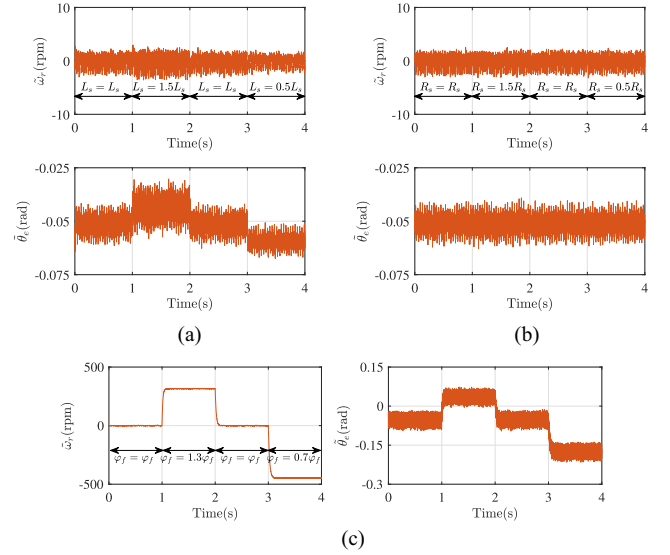


Fig. 17. Experimental results of the ONRM-REKF for step changes in parameters of the SPMSM. (a) L_s varies. (b) R_s varies. (c) φ_f varies.

1 Nm. Since the motor is in a reverse rotation state, a positive load value means it operates in generator mode. As shown, when the load changes, the speed estimation error and position estimation error converge to near zero after a brief dynamic process. And the motor remains stable and operates at the reference speed under different loads.

The parameter sensitivities of the proposed ONRM-REKF are also investigated at 1500 rpm in Fig. 17. Fig. 17(a) and (b) illustrate the performance of the ONRM-REKF for $\pm 50\%$ step changes in the stator resistance R_s and the stator inductance L_s of the SPMSM, respectively. These two parameters are primarily used in (3) to calculate the back EMF. It is clear that the variation of L_s has a slight impact on the estimation error of the ONRM-REKF, but it remains within an acceptable

range, while the variation of R_s has almost no impact. The flux linkage φ_f is only used to calculate the REKF. Fig. 17(c) shows that variations in φ_f significantly affect the steady-state error of speed estimation, which in turn impacts the steady-state error of position estimation. This phenomenon also exists in the conventional EKF and has been discussed and studied in [32] and [33]. The reason is that the EKF cannot distinguish the error of the flux linkage from the error of the estimated speed. When φ_f becomes a times its original value, $\hat{\omega}_e$ tends to become $1/a$ times its original value. Since $1/0.7 > 1/1.3$, the estimation error for $\varphi_f = 0.7\varphi_f$ in Fig. 17(c) is larger than that for $\varphi_f = 1.3\varphi_f$. In conclusion, the accuracy of φ_f must be ensured when using the REKF.

V. CONCLUSION

In this article, a novel sensorless control scheme has been proposed for PMSM drives considering noise and speed reversal. The proposed ONRM uses a newly defined objective function to address the case of speed reversal. The position octant method is proposed to ensure the convergence of NRM. Through the analysis with noise, the applicability condition of the ONRM is provided. Furthermore, the designed REKF filters the noise and estimates the speed. Numerous experimental results demonstrate that the proposed sensorless control scheme performs better compared to conventional PLL.

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